Rethinking Crowdsourcing Annotation: Partial Annotation With Salient Labels for Multilabel Aerial Image Classification

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***Abstract*— Annotated images are required for supervised model training and evaluation in aerial image classification. Manually annotating images is arduous and expensive, especially for aerial images, which often cover a large land area with multiple labels. A recent trend for conducting such annotation tasks is through crowdsourcing, where images are annotated by volunteers or paid workers [e.g., annotation volunteers for Open Street Map (OSM)] online from scratch. However, for crowdsourcing image annotations, the quality cannot be guaranteed, and incompleteness and incorrectness are two major concerns. To address such concerns, we have a rethinking of crowdsourcing annotations: our simple hypothesis is that if annotators only partially annotate multilabel images with salient labels they are confident in, there will be fewer annotation errors and annotators will spend less time on uncertain labels. As a pleasant surprise, with the same annotation budget, we show that a multilabel aerial image classifier supervised by images with salient annotations can outperform models supervised by fully annotated images. Our contributions are twofold: an active learning way is proposed to acquire salient labels for multilabel aerial images and a novel adaptive temperature associated model (ATAM) specifically using partial annotations is proposed for multilabel aerial image classification. When tested on practical crowdsourcing aerial data, the OSM dataset, the proposed ATAM can achieve higher accuracy than state-of-the-art classification methods trained on fully annotated images. The proposed idea is promising for crowdsourcing aerial image annotation. Our code will be publicly available.**

***Index Terms*— Aerial images, crowding sourcing, multilabel classification, salient labels.**

I. Introduction

With the increasing amount of images being available, manually annotating such images is a challenging task. For example, the well-known OpenImages test set will cost U.S. $6.5M for precise annotations, and only less than 1% OpenImages data are with annotations. Unannotated data are generally ignored when training a supervised learning model.

Generally, supervised or semisupervised models trained on more annotated data have better generalization on new data. Efficiently annotating unlabeled data is important, yet difficult, especially for multilabel scenes. For multilabel classification, one key challenge [1], [2] is how to make use of label correlations. A traditional way is ranking/retrieval-based approach [3], [4], in which the labels with the highest confidences are selected and assigned. Most recent methods are based on graph convolution network (GCN) and have two branches: the GCN branch is for creating category correlations and the CNN branch is for classification [5], [6]. Through the message propagation mechanism of GCN, spatial and topological relationships embedded in the edges of graphs are employed, to correlate multilabels. However, a potential concern is the annotation quality. Conventional multilabel aerial image classification methods focus more on mining the relationships between categories based on reliable labels [7], [8], while providing clean, multilabel annotations manually for large datasets has long been a challenging task.

The annotation task is even more challenging for multilabel aerial images, which naturally cover a relatively large land area and contain a large number of labels. A recent trend for aerial image annotations is resorting to crowdsourcing data, such as Open Street Map (OSM). OSM is an editable map that is built and annotated by volunteers from scratch. However, the quality of such a manually annotated map may not be satisfying. Incompleteness and incorrectness are two major concerns, as shown in examples in Fig. 1. Another famous crowdsourcing system is Amazon Mechanical Turk (AMT), where users (known as Taskmasters) submit their annotations for small payments. However, such annotations may not be reliable. There are also models specifically using crowdsourced remote sensing data. For example, in [9], crowdsourcing is used for detecting collapsed buildings, while in [10], crowdsourced data are used for monitoring lake motions. These applications show the potential and effectiveness of crowdsourcing in the remote sensing area. Therefore, finding an effective way for collecting crowdsourcing data is a promising research topic.

The quality of annotations is influenced by two main factors, i.e., the annotator's reliability [11]–[15] and the task's difficulty level [16]–[18]. Reliable annotators will assign labels seriously, while unreliable ones could assign labels carelessly in less than 10 s. The task's difficulty level is a major reason for label noises in crowdsourcing data. Since for a crowdsourcing platform, annotator's reliability is hard to control,

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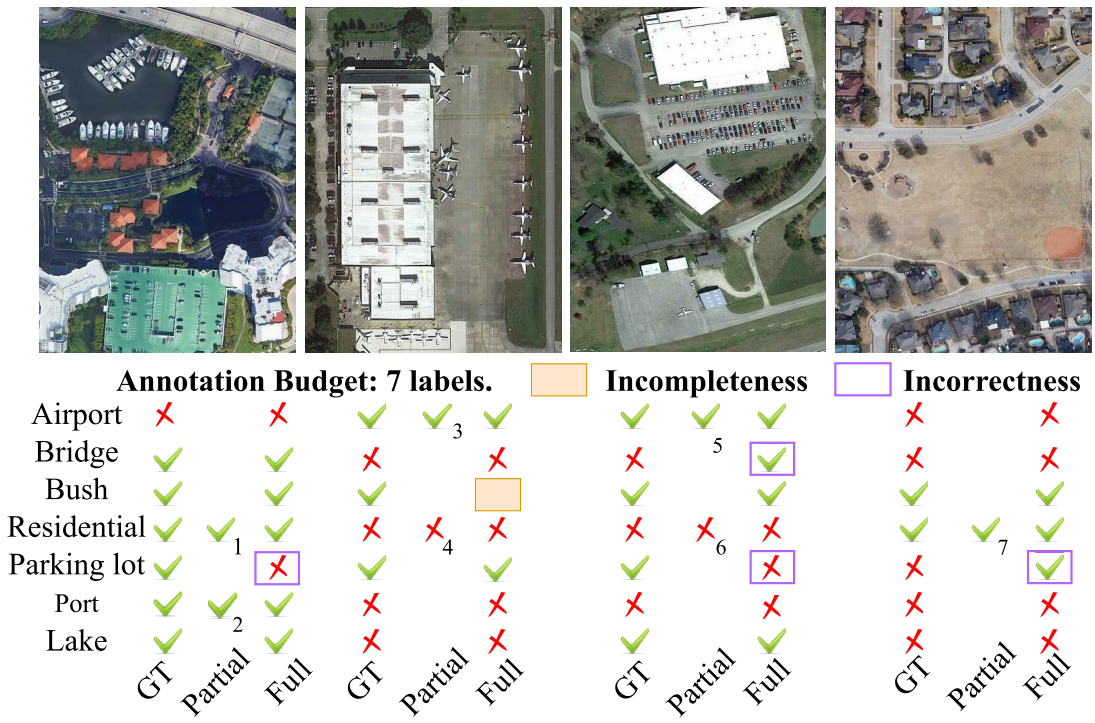


Fig. 1. Illustrative comparison of partial and full crowdsourcing aerial image annotations. GT means ground truth. Our rethinking is motivated by the following. With the same annotation budget (e.g., seven labels), we can fully annotate one image or instead partially annotate four images efficiently and confidently with salient labels. Since the full annotation task is more difficult and incompleteness and incorrectness problems are common, partial annotation may be a good alternative.

our rethinking of crowdsourcing annotation is mainly about the annotation task itself.

In this article, we have a rethinking of efficient and trustworthy crowdsourcing aerial image annotation. We believe that the annotators do not need to fully annotate the images with all labels as in the traditional way. It is preferred that they partially annotate images with the most salient labels. Annotating in this way can save the annotators the trouble of deciding uncertain labels. As they only annotate the labels they are confident in, the quality of annotations will be better and there will be fewer incompleteness/incorrectness cases. An example of partially annotating salient labels against full annotations is shown in Fig. 1. The major contributions of this article are given as follows.

1) We propose a partial crowdsourcing aerial image annotation idea. Volunteers/workers are required to annotate the assigned annotation budget (e.g., the total number of labels), instead of annotating the assigned number of images. Compared with the partial multilabel classification task, annotating in this customer-oriented way will have higher practical value. Also, training models with these customer-oriented annotations would be more challenging, as the data annotated in this way might suffer from problems such as annotation imbalance.
2) We propose a novel active learning approach for salient label sampling. We therefore for the first time create partially annotated datasets with salient labels.
3) We propose a novel partial annotation learning algorithm, the adaptive temperature associated model (ATAM), for multilabel aerial image classification.
4) We experimentally show that, given the same annotation budget, the proposed classification model trained on partially annotated aerial images outperforms the model trained with fully labeled aerial images.

II. Related Work

A. Crowdsourcing Image Annotation

Crowdsourcing services have facilitated researchers in collecting large-scale ground-truth labels [11], [19]. There are many crowdsourcing platforms, such as AMT, which has recruited more than 500k workers from 190 countries for annotation tasks [20], [21]. However, due to the openness of crowdsourcing, the annotators may yield low-quality annotations with noisy labels. Obtaining clean data with real ground-truth labels from the noisy labels remains a challenge [22]–[25]. Such a crowdsourcing strategy has two potential problems. First, different annotators will provide different service quality. For instance, an experienced annotator is more likely to provide high-quality annotations than an inexperienced annotator. Treating each annotator equally is not a wise way [12], [14], [26]–[28]. Second, typical crowdsourcing strategies (e.g., majority voting) are based on repeated annotations by multiple annotators to achieve higher annotation accuracy, which would be labor-expensive. In this article, to address the above concerns, we provide an alternative way for crowdsourcing image annotations: The annotators only need to partially assign the salient labels to images in which they are most confident. In this way, fewer annotators are needed for each task, and annotators are more likely to provide high-quality annotations.

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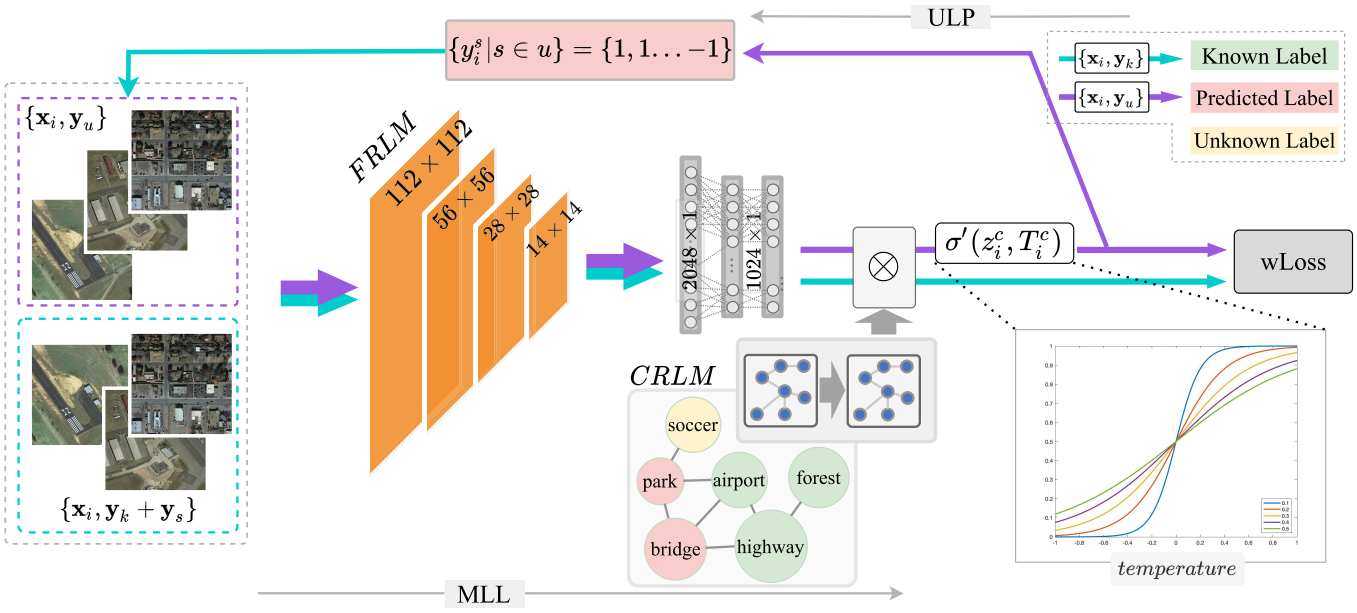


Fig. 2. Training model structure of the proposed ATAM framework. The framework uses ResNet as the backbone. The ATAM mainly includes two branches: the MLL branch and the ULP branch. The MLL has two main modules: the FRLM and the CRLM. Here, wLoss is the weighted loss in (7), and T is the smoothing factor.

B. Partial Multilabel Learning (PML)

Multilabel learning (MLL) has been an active topic with practical importance since images collected in the wild are often with more than one label [29]. The task of image MLL is to accurately recognize multiple objects or targets in an input image [30]. The conventional MLL research [31]–[35] mainly relies on the assumption that a small subset of images with full labels are available for training. Most recent MLL methods introduced adjacency-based labels, label-specific features, or new embedding in order to better explore label correlations [36]–[38]. The introduction of attention mechanism further increases the model generalizability of MLL and helps with the learning of semantic correlation [30]. However, the model training is still based on the fully annotated training subset, which is difficult to obtain in practice, as manual annotations always suffer from incomplete and incorrect annotation problems, as shown in Fig. 1. Such label concerns are especially serious for data annotated in a crowdsourcing way.

It would be much easier if crowdsourcing annotations are partial. With the emerging area of partial MLL, it becomes possible for researchers to learn a multilabel classification model from such ambiguous data [39]–[41]. Traditional methods treat missing labels as negatives during the model training process [42]–[47], while it may lead to classification performance degradation. To mitigate this problem, a novel approach for partial MLL treats missing labels as a hidden variable via probabilistic models and predicts missing labels via posterior inference [48], [49]. The work in [50] treats missing labels as negatives and then corrects the induced errors by learning a transformation on the output of the multilabel classifier. However, scaling the above models to large datasets is not easy [51], [52]. Another recent trend for partial MLL can be found in [53]–[60], which introduces curriculum learning and bootstrapping to increase the number of annotations. During model training, this approach uses partially annotated data and unannotated data whose labels the classifier is most confident with [61] and [62]. However, this way might introduce wrong labels that will taint the training data. This problem is called semantic drift. To better mitigate the semantic drift problem, in this article, we propose a novel image classification model using partially annotated data. Our motivation is to soften the strict discrepancy between negative and positive supervisory signals by introducing a novel smoothing factor, as in Fig. 2. To achieve this goal, we simultaneously learn image and label relationships. The proposed model is named the ATAM.

Algorithm 1 Active Learning for Annotation Sampling

Given: X_0 with annotation Y_0, the number of annotations N_t in iteration t, and the annotation budget L_b.
Initialize the pretrained network with parameter θ fine-tuned by Y_0.
while $L_b \geq \sum_{n=1}^{t} L_n$ **do**
  $Y_u^t = f(x_i; \theta)|_{i=1}^{N_t}$
  annotate labels in Y_u^t and update the Y_t with Eq. (1).
  Get the current number of known labels $\sum_{n=1}^{t} L_n$.
  Optimize θ with Y_t.
end while

III. Data Preparation

This section employs the active learning way to prepare salient labels for partially annotated multilabel aerial images.

We would like to clarify that partial annotations include both known and unknown labels. As shown in Fig. 3, the green area represents the known labels and the yellow area represents unknown labels. In the figure, we can note that at the image

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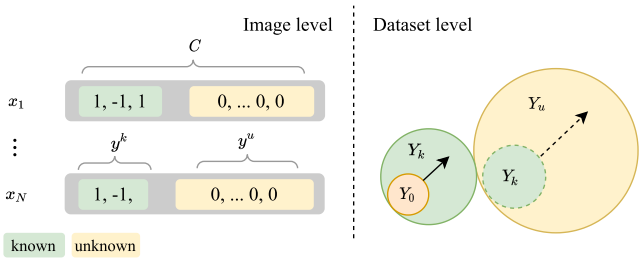


Fig. 3. Partial annotations at the image level and the dataset level. The arrow with Y_0 shows the procedure of active learning for data preparation. Y_k with the dashed arrow shows the procedure of ULP.

level, partial annotations include unknown labels (compared with full annotations). At the dataset level, partially annotated data are used as training data.

Given a training dataset with partial annotation $\{x_i, y_i^k, y_i^u\}_{i=1}^N$, y_i^k and y_i^u denote the known labels and the unknown labels of the input image x_i. Note that the total number of training images N is not a constant. Only if the volume of annotation set Y_k reaches the annotation budget L_b, the annotation process will stop. Suppose that we have C categories and $c \in C$ represents a specific category. For a deep network $f(\theta)$ (θ means network parameters), the predicted annotation $\hat{y}_{i_c} = f(x_i; \theta)$. $\hat{y}_{i_c} \in \{1, -1, 0\}$ stands for {*positive*, *negative*, and *unknown*} of category-c, respectively. Y_k and Y_u represent the known label set and unknown label set of the dataset, respectively.

For data preparation, we first annotate 50 aerial images collected from OSM with salient labels, represented by X_0. We use these annotated samples to fine-tune the backbone network (ResNet, as in Fig. 2), which is pretrained on the ImageNet training dataset. Then, iteratively, we input another 50 samples to the fine-tuned network and select the labels with the highest prediction confidence to annotate. We assume these salient labels that are also easiest to recognize by annotators. We further formulate this querying step as

$$Y_t = Y_{t-1} \cup \{f(\boldsymbol{x}_i; \theta) \geq S\}_{i=1}^{N_t} \tag{1}$$

where S is the confidence that a specific class exists (for positive labels), or does not exist (for negative labels), in sample i. We use this parameter for controlling the learning speed of active learning, as the number of labels to annotate in one iteration depends on the value of S. We empirically set it as 0.8. At the querying step t, a new subset N_t (e.g., 50) of samples will be actively annotated, and their annotations will be added to the current annotation set Y_t. The current annotated data will be further used to fine-tune the backbone network, which will be used for salient label querying in the next iteration. The query process will continue until the volume of the current annotation set Y_t reaches L_b. The detailed procedure is given in Algorithm 1. The data will be sampled and annotated iteratively in this querying-labeling way. Our goal is to generate the final annotation set Y_k (equals Y_t). Note that during this active learning process, we will ensure that at least one positive label is assigned for each image.

IV. Adaptive Temperature Associated Model

To efficiently utilize the annotation budget, we propose a novel algorithm, named the ATAM, for multilabel image classification using partially annotated data. The flowchart of the proposed ATAM model is shown in Fig. 2. The framework consists of two main branches: the MLL branch and unknown label prediction (ULP) branch.

A. Multilabel Learning

For the MLL branch, the goal is to optimize the framework using the annotations $\{Y_k \cup Y_s\}$ of the current step, where Y_k is the initial known labels and Y_s is the predicted labels generated by the ULP branch. The MLL branch consists of two modules: the feature representation learning module (FRLM) and the category representation learning module (CRLM). The FRLM module is used to extract the features of the input image x_i, which is a regular CNN as in Fig. 2. The CRLM module is introduced to integrate the category information into classification. The general operation routine of the procedure is as follows. A 2048×1 feature vector of the input data x_i is first extracted by the FRLM module and fed into two fully connected (FC) layers. Then, the output feature vector and the output of GCN (a $1024 \times C$ matrix) are fed to a scalar product layer, whose output is denoted as z_i. z_i is the input for the final activation layer σ, after which we obtain the final predicted labels y_i.

1) Category Representation Learning With GCN: In multilabel classification, inner correlations between the categories contain important information. To model the category-level representation, a GCN is introduced. GCN is intended to solve the problem under a non-Euclidean topological graph. A graph $\mathcal{G}(V, E)$ is with node features and edge features, where V defines a set of nodes and E is a set of edges. For the input of the category representation learning module, GloVe [63] is used to generate the category vectors. The computation graph is generated based on the feature embedding of each node and its neighbors. We follow a common practice to deploy the GCN layers:

$$H^{l+1} = \sigma\left(\hat{A} H^l W^l\right) \tag{2}$$

where H^l is the hidden representation describing the node's state at layer l for the input x_i, $\hat{A}$ is the normalized correlation matrix, W is a learnable transformation matrix, and σ acts as a nonlinear function, where we employ LeakyReLU to implement this operation. The correlation matrix $\hat{A}$ represents the co-occurrence of each two labels. For example, if one sample includes both labels i and j, then $A_{i,j}$ increases by 1. The final A will be normalized, to get $\hat{A}$, and the detailed illustration can be found in Algorithm 2.

B. Temperature Integrated ULP

The problem we tackle in the ULP branch is to assign full annotation to the partially annotated training samples, under the limitation of the annotation budget. To be more specific, the ULP branch will generate Y_s (as mentioned earlier) for the unannotated part of the training samples.

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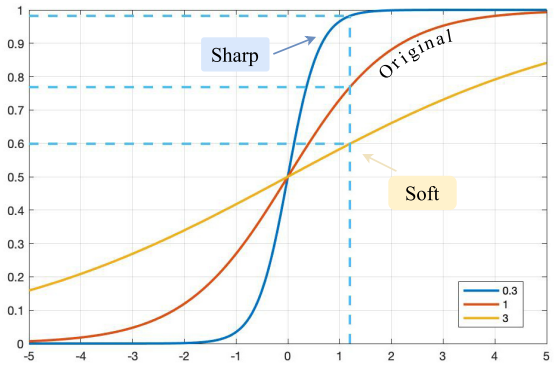


Fig. 4. Illustration of how the smoothing factor T works.

Given the output of the last activation layer in MLL $p_i = \{p_i^k, p_i^u\}$ representing the possibilities of occurrence of known labels and unknown labels, the procedure of selecting y_i^s could be formulated as

$$y_i^s = -\mathbb{1}\left[p_i^u < (1-\beta)\right] \cup \mathbb{1}\left[p_i^u \geq \beta\right]. \tag{3}$$

Here, we empirically set β as 0.8 and start to generate y_s after around ten epochs of training. The newly generated y_s in each epoch is regarded as pseudo ground truth and combined with Y_k as $\{Y_k \cup Y_s\}$, which will be used for the following training process.

There exist two problems in the classification procedure. First, we need to improve the confidence level of Y_s as the output Y_s at the very beginning is not valid enough. Second, there must be some difficult labels whose occurrence possibilities will always stay in $[1-\beta, \beta]$, which means that we cannot classify them as positive or negative.

Inspired by the knowledge distillation idea, we propose to introduce the temperature concept to sharpen/soften the invalid/difficult label predictions. To be more specific, we introduce a temperature factor T to the sigmoid function. Integrated with the temperature factor T, for a random sample i and category c, the sigmoid function (activation) could be defined as

$$\sigma'\left(z_i^c, T_i^c\right) = \frac{1}{1+\exp\left(-z_i^c / T_i^c\right)} \tag{4}$$

Algorithm 2 Correlation Matrix Construction $\hat{A}$

Input: $\boldsymbol{y} = \{y_c, y_s\}$.
Output: normalized correlation matrix.
while epoch $\leq$ max epoch **do**
  for $idx_i, idx_j \leftarrow 1$ **to** num_categories
    if $\boldsymbol{y}[idx_i]$ & $\boldsymbol{y}[idx_j] \neq 0$ **then**
      cor_count$[idx_i][idx_j]$ += 1
  end if
  end For
  for idx $\leftarrow 1$ **to** num_categories **do**
    cor_count$[idx] = \frac{cor_count[idx]}{\sum cor_count[idx][:]}$
  End For
End While

where z is the input of the activation layer in the neural network, which is the dot product between outputs of CRLM and FRLM. To calculate T, we will introduce the edge weight in GCN as in Section IV-A.

We assume that an unknown label m is a difficult label if its connection $E_{m,1}, E_{m,2}, \ldots, E_{m,n}$ (suppose that we have n known labels) with all known labels are small, which means that CRLM cannot help with the prediction. Then, we will sharpen the prediction. If the connection between $E_{m,k}$ m and a random known label k is too large, then the prediction relies heavily on CRLM, which means that m might have the same annotation as k, and invalid prediction might be given at the very beginning of model training. Then, we need to soften this prediction. We can also understand this assumption as we want to use T to better balance the effectiveness of CRLM and FRLM. Therefore, we define T as

$$T_m^c = \alpha \cdot \mathrm{STD}\left(E_{mj}\right)|_{j=1}^n \tag{5}$$

where STD means the standard deviation and α is a hyperparameter for controlling the overall smoothing level and is empirically set to be 10. A visualized function of T can be found in Fig. 4. We also want to point out that we set the maximum training epochs of ULP as 50. After 50 epochs, there are still a few difficult labels (1%–2%). We find that the annotations of such difficult labels do not make a big difference for the following model training, and we set all these labels as negative labels.

C. Weighted Focal Loss and Label Smoothing

An inherent problem for multilabel classification is label imbalance. There are much fewer positive samples than the negative samples. To overcome this problem, we use the weighted focal loss (wFL) instead of the regular binary cross-entropy loss for the optimization of the proposed model, which is formulated as

$$\begin{aligned}\mathcal{L}_k = -\sum_{i=1}^{K} p_\beta \big(&\alpha y_i^k (1-\sigma(\cdot))^\gamma \log \sigma(\cdot) \\ &+ (1-\alpha)\left(1-y_i^k\right)\sigma^\gamma(\cdot)\log(1-\sigma(\cdot))\big)\end{aligned} \tag{6}$$

where p_β is the proportion of classwise samples with respect to all the data in the dataset and fulfills $p_\beta \in (0,1)$& $\sum_{\beta=1}^{C} p_\beta = 1$. α and γ here are empirically set as 0.25 and 2, respectively, as demonstrated in [64].

To calculate the loss $\mathcal{L}_s$ for the predicted labels Y_s (Y_s is also taken as ground truth in the later stage of training), we introduce the smoothing factor T to the activation function σ. By changing σ in (6) to σ, we can get $\mathcal{L}_s$.

We further use a hyperparameter ϵ to control the ratio of $\mathcal{L}_k$ and $\mathcal{L}_s$. We give higher priority to Y_k and ϵ is empirically set as 0.5. The final loss for training is formulated as follows:

$$\mathcal{L}_w = \mathcal{L}_k + \epsilon \mathcal{L}_s. \tag{7}$$

D. Model Training

Note that the ULP branch is not trainable, as the only variable T in ULP depends on the GCN part in the MLL branch.

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Algorithm 3 Training for ATAM

Input: Training set $\{x_i, \boldsymbol{y}_i^c, \boldsymbol{y}_i^u\}_{i=1}^{N_t}$, known label set Y_k, unknown label set Y_u.
Output: optimized θ
{// Step 1: Initial the network using Y_k }
while (stopping criteria)
  Update θ with Y_k by minimizing $\mathcal{L}_{known}$
end While
Select Y_s from Y_u according to Eq. (3).
$Y_k \leftarrow \{Y_k \cup Y_s\}$; $Y_u \leftarrow \{Y_u \backslash Y_s\}$
{// Step 2: Unknown label prediction.}
repeat
  while (stopping criteria)
    Update θ with Y_k by minimizing $\mathcal{L}_w$ {// PML branch}
  End while
  Select Y_s from Y_u according to Eq. (3). {// ULP branch}
  $Y_k \leftarrow \{Y_k \cup Y_s\}$; $Y_u \leftarrow \{Y_u \backslash Y_s\}$
until stopping criteria
$Y_u \leftarrow negative$ {// Step 3: Multi label classification with full label.}
while (stopping criteria)
  Update θ with $\{Y_k \cup Y_u\}$ by minimizing $\mathcal{L}_w$
end while

The training is for the MLL branch. After getting the known label set Y_k using active learning, as described in Section III (step-1), we first use the known labels (x_i, y_i^k) to train the MLL branch by minimizing the loss function $\mathcal{L}_k$ until we get the first predicted annotations Y_s (around ten epochs). Second (step-2), with the optimized MLL branch, we get the new label predictions. We use (3) to obtain the predicted annotations Y_s and extend the training annotations to $(Y_k \cup Y_s)$. Third (step-3), the MLL branch is further optimized by minimizing $\mathcal{L}_w$ using the updated training dataset $(Y_k \cup Y_s)$. By repeating step-2 and step-3, the annotations of all categories will be acquired gradually, and the network parameter θ for the MLL branch will be optimized. After around 50 epochs, all labels for training data are acquired, and the ULP branch will be removed. The following training is the regular network training for the MLL branch until convergence. For model testing, we also only use the MLL branch to obtain the multilabel prediction of each testing sample. The steps for training the framework are shown in Algorithm 3.

V. Experiments

A. Datasets, Setup, and Evaluation Metrics

To evaluate the performance of the proposed method, we conduct experiments on the crowdsourcing OSM dataset. OSM as an aerial view image data covers a large area of interest and is more difficult for classification. In our dataset, as we partially annotate the OSM with the labels from the AID dataset, the created dataset is further named OSM-AID. We will release this OSM-AID data in our project website.

OSM-AID: OSM with AID-label annotations has 20 classes, including Airport, Baseball, Beach, Bridge, Commercial, Farmland, Forest, Lake, Parking Lot, Park, Port, Residential, River, Roundabout, Soccer Field, Stadium, Storage Tank, Train Station, Works, and Sea. The dataset has 3234 images in total, with 64 680 labels, and the images in each class are imbalanced, ranging from 200 to 700. The labels are annotated in a crowdsourcing way by our lab and will be released. The input images are rescaled to 512×512. The images are split into training/testing in a ratio of 8:2, more specifically, 2580 as training (2100 for training and 480 for validation) and 654 as testing.

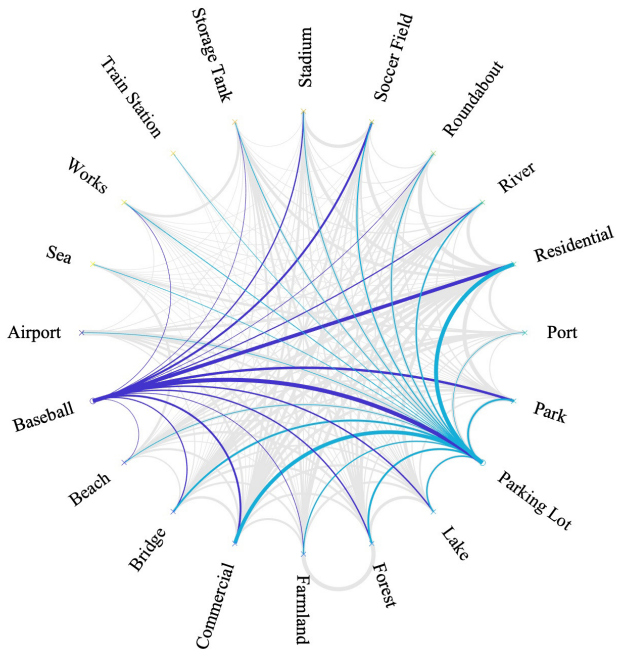


Fig. 5. Label correlations of the OSM-AID dataset.

Since the OSM-AID dataset is created by ourselves, we have shown examples in Fig. 6.

To estimate the relationships between the annotations, we calculate the label correlation matrix A, as we mentioned in GCN. Using labels in the training dataset, we count the times of the occurrence of different label pairs. By visualizing the matrix A of the OSM-AID dataset, as shown in Fig. 5, we note that the correlations between the labels have some patterns. For example, there is "parking lot" near the "residential" or "commercial." This co-occurrence should be considered in the modeling part, and we believe that the introduction of this A and the CRLM should be necessary.

Regarding the model setup, ResNet-101 [65] is used as the backbone and is pretrained on ImageNet [66]. For different datasets, the input images are randomly cropped and resized for data augmentation. SGD is used for network optimization. The momentum is set to be 0.9, with a decay of 10^{-4}. The batch size is 12, with the input size 512×512. The learning rate starts at 0.01 and decays by a factor of 10 for every 50 epochs. The model is trained on Google Colab with Nvidia Tesla P100.

For performance evaluation, we report the average overall precision (OP), overall recall (OR), F1-score (OF1), and F2-score (OF2).

B. Performance Comparison

In this section, we conduct several experiments to explore the best strategy to utilize the annotations with a specific budget. For each experiment, the annotation budget is the same among different methods.

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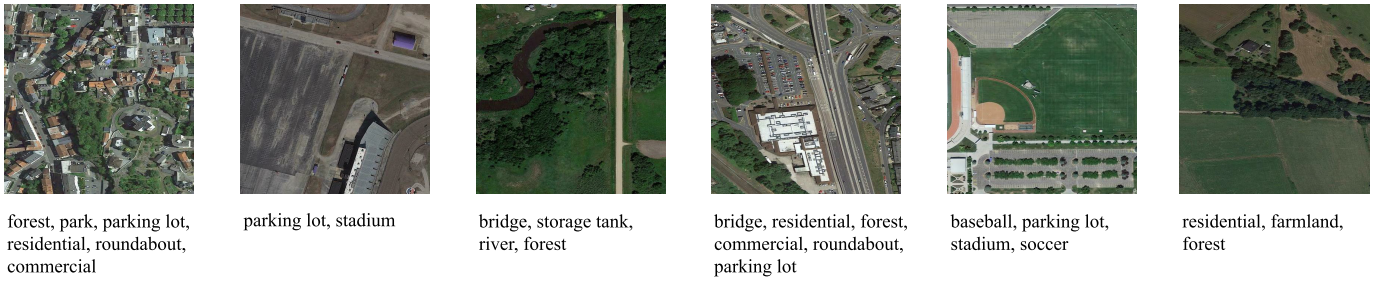


Fig. 6. Example images with a different number of categories from the OSM-AID dataset.

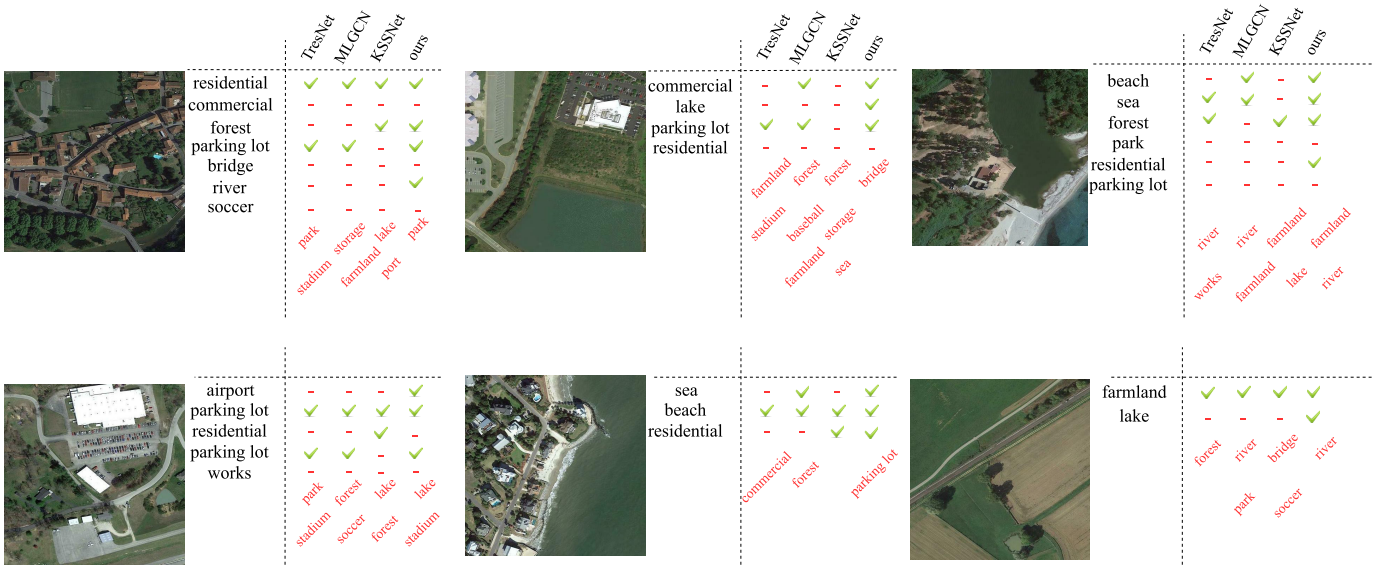


Fig. 7. OSM-AID: visualization examples of classification outputs using TresNet (N-ML), KSSNet (N-ML), ML-GCN (N-ML), and the proposed method. Here, N-ML means negative missing labels.

TABLE I

CLASSIFICATION ACCURACY COMPARISONS WITH SOTA METHODS (I.E., TRESNET-ASL [67], ML-GCN [68], KSSNET [69], Q2L-R101 [70], AND P-ASL [71]) AND DIFFERENT ANNOTATION STRATEGIES

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
TresNet-ASL [67]	0.5700	0.3701	0.4488	0.3980	0.5753	0.3636	0.4456	0.3925
ML-GCN [68]	0.4620	0.4034	0.4307	0.4139	0.6800	0.3508	0.4629	0.3884
KSSNet [69]	0.4704	0.3622	0.4093	0.3797	0.5231	0.3552	0.4231	0.3796
Q2L-R101 [70]	0.6237	0.4125	0.4966	0.4425	0.6308	0.4018	0.4909	0.4333
P-ASL [71]	0.5691	**0.4712**	0.5155	0.4880	0.5902	**0.4632**	0.5190	**0.4840**
ATAM	**0.6826**	0.4567	**0.5472**	**0.4891**	**0.6935**	0.4497	**0.5456**	0.4837

TABLE II

EXTENDED CLASSIFICATION ACCURACY COMPARISONS WHEN INCORRECT NEGATIVE LABELS EXIST

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
TresNet-ASL(N-ML) [67]	0.4154	0.2053	0.2748	0.2284	0.5888	0.1890	0.2862	0.2187
ML-GCN(N-ML) [68]	0.6629	0.1277	0.2142	0.1523	0.6627	0.1371	0.2272	0.1629
KSSNet(N-ML) [69]	0.5149	0.1629	0.2475	0.1887	0.6630	0.1601	0.2579	0.1887
ATAM	**0.6826**	**0.4567**	**0.5472**	**0.4891**	**0.6935**	**0.4497**	**0.5456**	**0.4837**

40% Training Data With Full Annotations Versus All Training Data With 40% Partial Annotations: In this experiment, three most recently proposed well-known multilabel classification algorithms, TresNet-448 [67], ML-GCN [68], and KSSNet [69], are used for comparison. As these state-of-the-art (SOTA) methods do not have a missing label prediction

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TABLE III
EXTENDED CLASSIFICATION ACCURACY COMPARISONS BETWEEN DIFFERENT ANNOTATION WAYS WITH TOP 3 PERFORMANCE

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
ATAM	**0.6826**	**0.4567**	**0.5472**	**0.4891**	**0.6935**	**0.4497**	**0.5456**	**0.4837**
ATAM(40-full)	0.5044	0.3723	0.4284	0.3929	0.5015	0.3701	0.4259	0.3906
ATAM(100-full)	0.7331	0.4850	0.5838	0.5202	0.7498	0.4715	0.5789	0.5093

module, they can only be trained with fully annotated data. For our ATAM method, we use partially annotated data for training.

All comparison methods have the same amount of annotation budget for model training. To be more specific, during the implementation, we use 40% partial annotations (16 800 labels) of training data to complete our experiments. We will also compare the influence of different proportions in the following.

The results are shown in Table I. With the same annotation budget, the proposed method outperforms other multilabel methods. The reason for the better performance is that with a fixed annotation budget, the partial annotation method could access 2.5 times the number of images compared with fully annotated data.

With the training process going on, the proposed method is able to predict missing labels step by step and further use the extended annotations to optimize the model. As can be found in Table I, the proposed method using partial annotations can achieve much higher accuracy than comparative methods using the fully annotated option.

Simulation for Data With Incorrect Labels: In this section, we only use partially annotated data. We want to investigate whether wrong labels exist, how accuracy changes. We assign all missing labels as negative values, the same as the crowdsourcing process, which generally takes these unannotated labels as negatives. This is also a common strategy for multilabel classification methods with missing labels. The same setting goes for both the proposed and the comparison methods. The comparison results are shown in Table II and a qualitative comparison is demonstrated in Fig. 7.

Full Versus Partial Annotations: We also use fully annotated data with the same annotation budget for the proposed ATAM method, to more specifically verify the effectiveness of partial annotation. Results can be found in Table I. We can find that with the same annotation budget, the partial annotation way significantly outperforms the full annotation way [ATAM(40-full)]. Furthermore, we use a full annotation budget (100% labels) instead of 40% to make a comparison with our partial method. Under such a scenario, the inputs of both methods are all the training images. We can find that compared with training on the whole dataset with full annotations [ATAM(100-full)], our classification accuracy only slightly degrades, though it saves 60% annotation budget. The detailed results can be found in Table III.

For SOTA methods, compared with using less but fully annotated images as in Table I, the corresponding F1-score and F2-score degrade significantly. One main reason lies in the fact that there are quite many positive labels in the missing labels. Assigning missing annotations with negative values will introduce too much noise to the training dataset. Also, it exacerbates the data imbalance problem of the multilabel dataset, as we mentioned before.

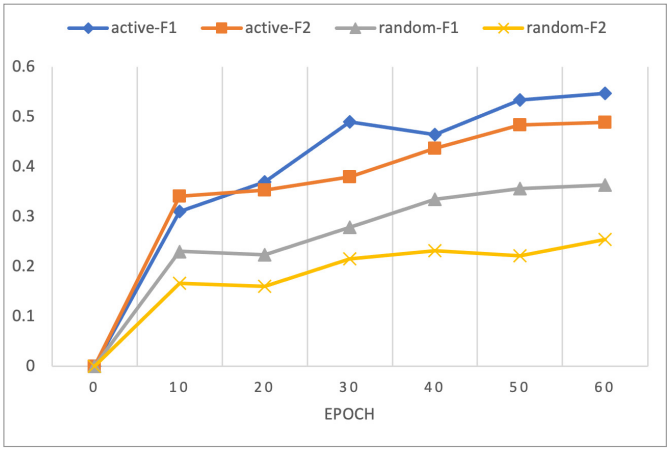


Fig. 8. Random annotation versus salient annotation.

C. Effects of Salient Sampling

In this section, we investigate the effect of active learning based on salient label sampling. In the labeling process, people always tend to provide annotations to those "easy" salient objects, especially when there is no restriction on which categories must be annotated. We want to verify the effectiveness of the salient annotation way and compare this way and the random annotation way in this section. Salient annotation is realized through active learning, as mentioned in Section III. For random sampling, 40% annotations are chosen randomly, including both positive and negative labels. For each sampled image, at least one positive label is chosen to avoid the annotated labels being all negative. For active learning, the initial 50 images use the same strategy as the random sampling; a querying-labeling way is applied to generate the rest labels.

As shown in Fig. 8, salient sampling can not only accelerate the convergence speed but generates better performance compared with random sampling as well. In other words, initializing the annotations properly could help the framework to work much more efficiently.

D. Effect of Label Proportion

We explore the effect of the proportion of known labels on the performance of the proposed method on the OSM-AID dataset. As shown in Fig. 9, with the label proportion increase from 40% to 80%, the F1-score increases from 0.5472 to 0.5801 and F2-score increases from 0.4891 to 0.5197. With

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TABLE IV
EXTENDED RESULTS WITH DIFFERENT CATEGORY EMBEDDING STRATEGIES

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
Word2vec [72]	0.6778	0.4625	0.5498	0.4939	0.6817	0.4502	0.5423	0.4830
fastText [73]	0.6525	0.4511	0.5334	0.4808	0.6598	0.4401	0.5280	0.4715
GloVe [63]	0.6826	0.4567	0.5472	0.4891	0.6935	0.4497	0.5456	0.4837

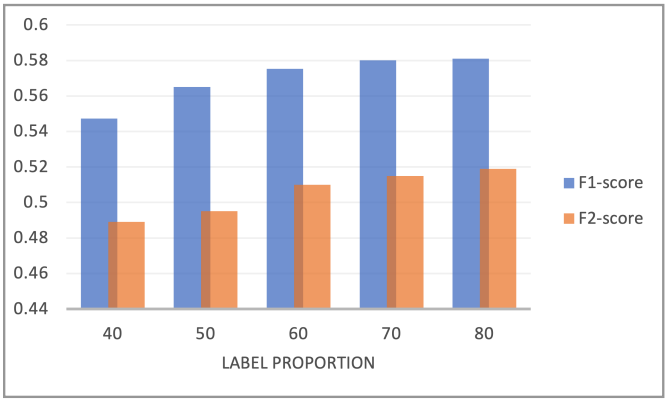


Fig. 9. Performance comparisons with different label proportions.

TABLE V
EXTENDED RESULT WITH DIFFERENT LOSS FUNCTIONS

Method	All			
	OP	OR	OF1	OF2
BCE	0.6630	0.2475	0.3604	0.2830
wLoss	**0.6826**	**0.4567**	**0.5472**	**0.4891**
	Top-3			
BCE	0.6783	0.2248	0.3377	0.2595
wLoss	**0.6935**	**0.4497**	**0.5456**	**0.4837**

a lower annotation budget, the proposed method could get a comparable accuracy as training with fully annotated data. We can conclude that annotation proportion around 50% should be preferred, as the labor wasted on more annotations cannot generate a correspondingly large increase in classification accuracy.

E. More Ablation Studies

1) Different Feature Representations: In this article, we use GloVe as a feature extractor. GloVe is used to obtain vector representations for categories. The input of the graph is word embedding generated by GloVe, and the node-level output feature (or intermediate feature) of GloVe is the prediction score of each node (category). Since the feature representation for MLL is important, we have done a separate ablation study for different feature representations, as in Table IV. We could see that the two embedding strategies, GloVe and Word2vec, have comparable results, better than that of fastText. Due to the fact that fastText operates at a more granular level with character n-grams, while GloVe and Word2Vec use words to predict words, the result is reasonable. Therefore, in our method, we use GloVe for feature extraction.

TABLE VI
EXTENDED RESULTS WITH DIFFERENT TEMPERATURE FACTORS

Method	All			
	OP	OR	OF1	OF2
T=1	0.6212	0.2771	0.3832	0.3116
T(ATAM)	**0.6826**	**0.4567**	**0.5472**	**0.4891**
	Top-3			
T=1	0.6278	0.2705	0.3781	0.3052
T(ATAM)	**0.6935**	**0.4497**	**0.5456**	**0.4837**

2) Ablation Study for Model Loss: We also compare our proposed wLoss with traditional BCE loss, to verify its effectiveness. The comparison can be found in Table V.

3) Ablation Study for the Introduced Temperature: To verify the effectiveness of the temperature, we have done an ablation study. We can note that, as shown in Table VI, with the temperature the performance of the proposed method increases significantly.

F. More Experiments on OSM-UCM Dataset

To verify the extensionability of the proposed model, we also create another new aerial dataset, a multilabel OSM-UCM dataset by combining the OSM images and UCM [2] labels. The OSM-UCM is a quite small dataset, containing 1799 images and 17 categories, and contains more noisy labels. Compared with the SOTA methods, as shown in Table VII, our method consistently yields the best performance. However, due to the noise, the performances of different methods on this dataset are generally worse than that on the OSM-AID dataset. Further improvements from both the dataset and algorithm aspects are needed for this dataset.

VI. CONCLUSION

In this article, inspired by our observation that partially annotating the crowdsourcing aerial images with salient labels can decrease the annotation task difficulty and increase the annotator's reliability, we propose a novel way for crowdsourcing aerial image annotation. We also propose a novel framework for multilabel image classification using partially annotated aerial images. The proposed ATAM can utilize the annotations more efficiently. In the experimental part, we demonstrate that, with the same annotation budget, the classification model trained with partially annotated data yields better performance. We will extend the proposed annotation framework to more image types and learning tasks in future work (a simple extension to Coco 2014 dataset can be found in the discussion).

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TABLE VII

EXTENDED RESULTS WITH TOP-3 PERFORMANCE ON OSM-UCM: CLASSIFICATION ACCURACY COMPARISONS WHEN INCORRECT NEGATIVE LABELS EXIST

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
TresNet-ASL(N-ML) [67]	0.2973	0.3031	0.3002	0.3019	0.2991	0.3125	0.3008	0.3018
ML-GCN(N-ML) [68]	0.3281	0.2877	0.3066	0.2950	0.3325	0.2968	0.3136	0.3033
KSSNet(N-ML) [69]	0.2722	0.2828	0.2774	0.2806	0.2833	0.2905	0.2869	0.2890
ATAM	**0.3259**	**0.3112**	**0.3184**	**0.3140**	**0.3301**	**0.3073**	**0.3183**	**0.3116**

TABLE VIII

EXTENDED RESULTS WITH TOP-3 PERFORMANCE ON COCO2014: CLASSIFICATION ACCURACY COMPARISONS WHEN INCORRECT NEGATIVE LABELS EXIST

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
TresNet-ASL(N-ML) [67]	0.5976	0.2917	0.3920	0.3250	0.6276	0.2744	0.3819	0.3092
ML-GCN(N-ML) [68]	0.5139	0.2236	0.3116	0.2521	0.6165	0.2138	0.3175	0.2459
KSSNet(N-ML) [69]	0.4696	0.2153	0.2952	0.2415	0.6082	0.2068	0.3087	0.2382
ATAM	**0.7853**	**0.5978**	**0.6788**	**0.6278**	**0.7901**	**0.5915**	**0.6765**	**0.6228**

VII. DISCUSSION

We also try to extend the proposed model to more scenarios, e.g., multilabel data Coco 2014. We want to further evaluate our proposed method in a real scenario when there are absent labels.

COCO2014 is mostly used in objection detection, containing 80 categories in total, 82 081 images for training and 40 504 images for validation and testing. As the dataset is not specifically for multilabel classification, the average number of categories in one image is only 3.5. With only 40% budget, it is hard to generate a partial annotation dataset fairly for different methods. We therefore randomly choose 60% labels (including originally negative ones) as missing labels and set them as "negative." In this way, we simulate the crowdsourcing data with noises (incorrect labels). We can note the performance superiority of our proposed method in Table VIII. This observation shows that the proposed framework is more robust when incorrect labels exist.

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